Numerical Modeling and Optimization of Electrode Induction Melting for Inert Gas Atomization (EIGA)

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Electrode Induction Melting Inert Gas Atomization (EIGA) is the state-of-the-art process for the high-quality spherical powder production for additive manufacturing needs. The growing demand for EIGA powders drives the interest for the scale-up of well-established atomization of small Ø50 mm Ti-6Al-4V electrodes, as well as atomization of new refractory materials like Tantalum. However, during first tests with Ø150 mm Ti-6Al-4V and Ø50 mm Tantalum electrodes, the difficulties with melting stability were observed. In order to overcome these difficulties and to improve understanding of details of inductive coupling and favorable melting conditions, a numerical model for the electrode induction melting has been developed and applied.

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I. INTRODUCTION

IN recent years, the demand for metal powders has grown significantly due to remarkable breakthroughs in 3D metal printing and other additive manufacturing (AM) technologies.[1] AM requires spherical powders with good rheological flow characteristics, and promises the manufacture of complex structures and parts with minimized raw material usage and at a lower cost compared with conventional manufacturing routes.

The Electrode Induction Melting Inert Gas Atomization (EIGA) process was developed and patented by ALD[2] as an alternative, ceramic-free melting and atomizing process that is especially suited for production of high-purity, reactive and refractory metal powders. Today EIGA is one of the leading processes for manufacturing Titanium, Zirconium, Niobium, and precious metal alloy high-quality spherical powders for aerospace, medical, energy, chemical, electronic, and other industry applications.[3]

In the EIGA process, the pre-alloyed cylindrical electrode is mounted on an electrode feeding device which continuously lowers the electrode into a conical induction coil (Figure 1). Then, energy is coupled into the electrode using a high-frequency electromagnetic (EM) field. In order to achieve more uniform (axisymmetric) melting, the electrode is slowly rotating around its symmetry axis (1 to 10 rpm). As a result, a melt film is formed on the electrode surface and a molten metal stream or droplets fall from the electrode tip into the inert gas nozzle where a high-velocity gas stream atomizes the melt.[4] By this means, the generated micro-droplets solidify while traveling down in the atomization tower and form spherical-shaped fine powders which are collected in a vacuum-tight powder container.

Induction coil design and melting parameters determine the size and superheat of liquid metal droplets that fall in the nozzle and therefore have a strong impact on the atomized powder characteristics. At this point, the process simulation could improve understanding of details of inductive coupling and favorable melting conditions.

Numerical modeling of metallurgical applications is generally multi-physical and involves coupling between fluid flow, electromagnetic, heat transfer, and other disciplines of physics. Apart from that, different numerical methods are beneficial for different types of problems, for example, Finite Element Method (FEM) is preferred in structural mechanics and electromagnetics, while Finite Volume Method (FVM) is widely applied for computational fluid dynamics (CFD).

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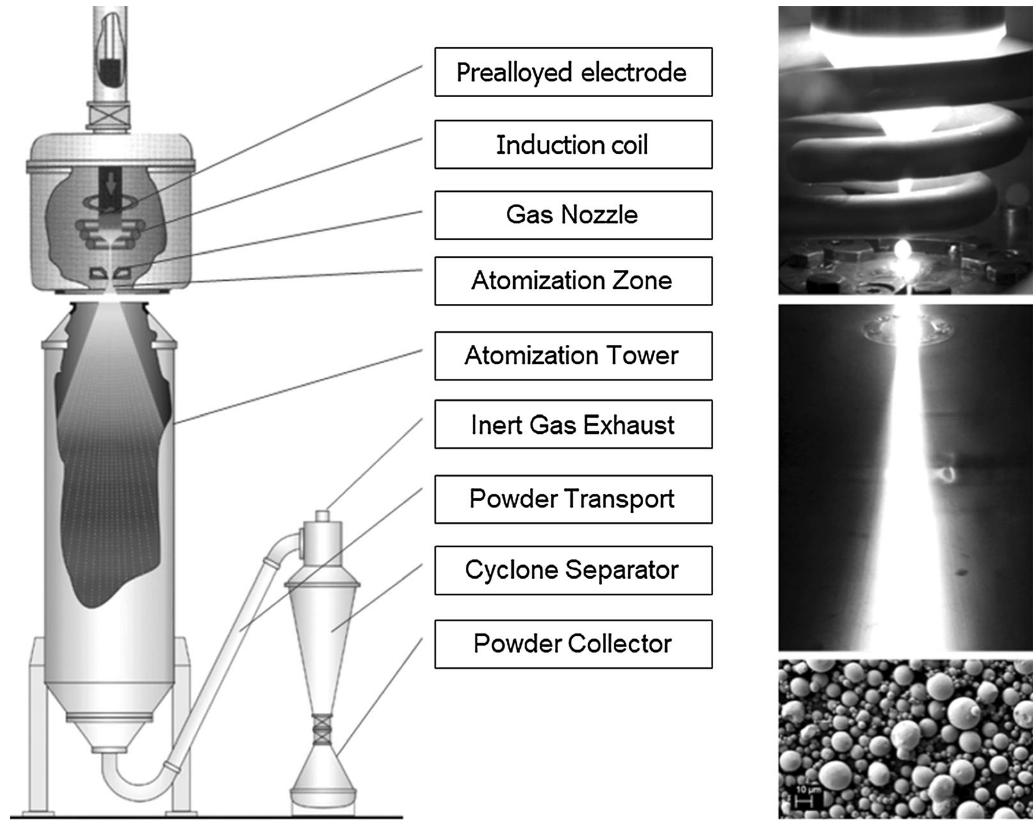


Fig. 1—The EIGA process and well-established atomization of Ø50 mm Ti-6Al-4V electrode.

However, it is possible to solve multi-physical metallurgical problems within a single commercial FVM simulation framework[5] or a self-written FEM software.[6] Alternatively, efficient models are developed by coupling between the finite element and the finite volume open-source frameworks.[7]

Numerical models for the melting dynamics in the EIGA process have been developed and applied to study the optimized EM coupling between induction coil and unmolten electrode tip,[8] favorable melt outflow, stability and temperature up to the moment of droplet detachment,[9] and laminar necking of droplets (still without detachment) and parameter influence on the melt superheat.[10]

However, there is no numerical model found in the literature that describes the process starting from preheating and initial melting of the electrode, formation of melt stream at the tip, droplet detachment, and steady-state dripping. The stable start of melting without clogging of the nozzle and the size and superheat of falling droplets during steady-state melting are both of great interest.

Later on, linking such numerical tool with advanced atomization models like[11] would allow to simulate a complete production route from the electrode to the powder and quantify the influence of melting conditions on the powder characteristics.

Meanwhile, the increasing demand for the EIGA powder drives the interest for the growth of productivity by scaling-up existing systems (atomization of larger diameter Ti-6Al-4V electrodes, *e.g.*, Ø150 mm), as well as atomization of new refractory materials (*e.g.*, Tantalum with especially high melting temperature of 3017 °C).

However, straightforward extension of EIGA applications has faced a problem: unusual electrode tip melting and growth of lamellae structures that remain cold, elongate, and finally build up to physically contact and short-circuit the induction coil (Figure 2). Increase of the electrode rotation aimed on achieving more axisymmetric power input and azimuthally uniform electrode melting did not alleviate the striation problem. Especially in the case of Tantalum, the radiation loss is so large that insufficient superheat

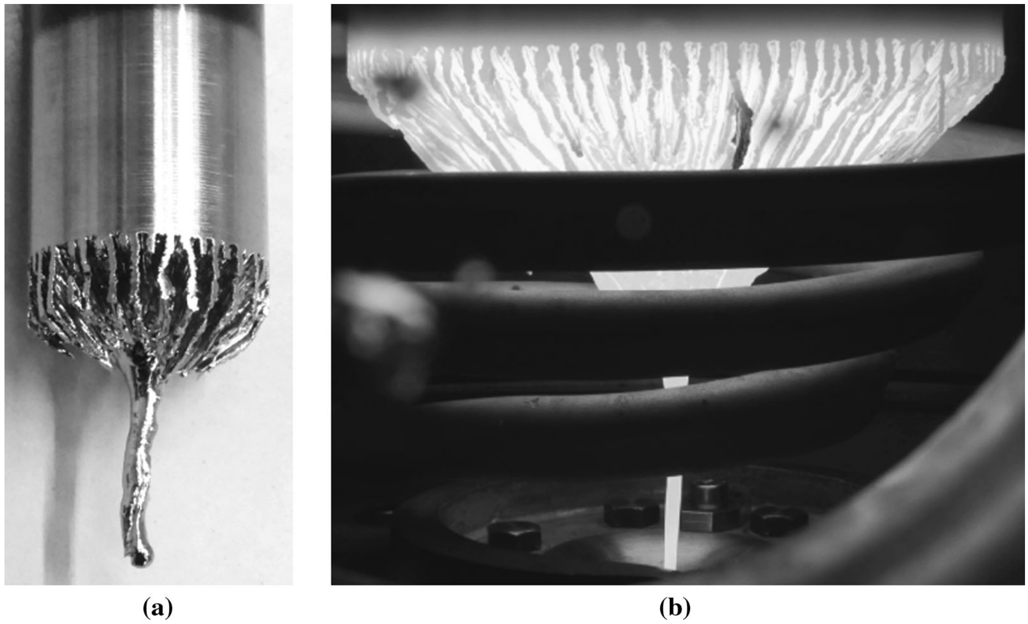


Fig. 2—Lamellae growth during melting of Ø50 mm Tantalum electrode (*a*) and Ø150 mm Ti-6Al-4V electrode (*b*).

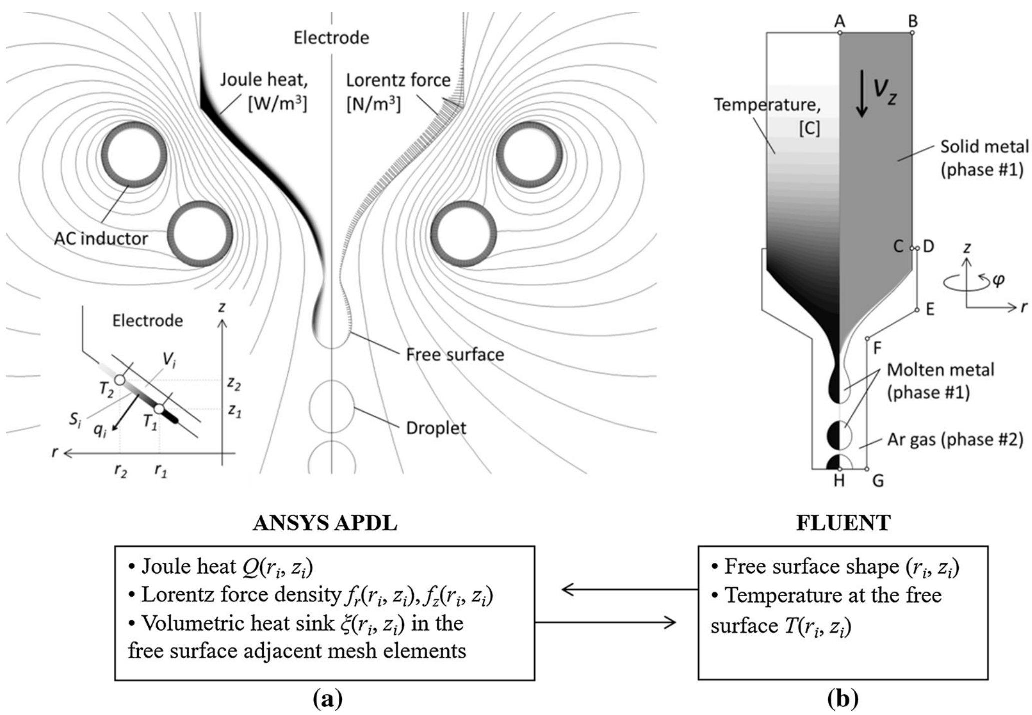


Fig. 3—Relationship of EM calculation in ANSYS (*a*) with the fluid flow and heat transfer calculation in FLUENT (*b*) *via* information exchange between the simulation programs.

results in re-solidification of the melt on its way to the nozzle. Therefore, optimization of melting conditions and re-design of induction coils had to be performed.

A numerical model for the electrode induction melting has been developed and applied to study the EIGA process and to solve the encountered melting issues.

II. NUMERICAL MODEL

In this work, a 2D-axisymmetric model for the molten metal flow with free surface in an alternate EM field has been developed by means of transient two-phase volume of fluid (VOF) turbulent flow simulation in FLUENT (FVM) and frequent Joule heat and Lorentz force recalculation in ANSYS APDL (FEM) (Figure 3).

Initial shape of the electrode, as well as every instant free surface shape obtained with two-phase flow calculation, is written to a text file. This file contains free surface keypoint numbers n_i, keypoint coordinates (r_i, z_i), and series of keypoint number sequences that indicate the order of free surface keypoint connection with lines for definition of 2D surface elements. Temperature $T(r_i, z_i)$ at the surface of the electrode is used for a simplified calculation of the heat radiation and is written to a text file as well.

The radiation heat flux q_i from the electrode free surface element located inside the calculation domain (Figure 3(a)) is defined as

$$q_i = \varepsilon \cdot \sigma_{\mathrm{StB}} \cdot S_i \cdot \left(T_\infty^4 - T_i^4\right) [\mathrm{W}], \quad [1]$$

where ε is emissivity, σ_{StB} is the Stefan–Boltzmann constant, S_i is the area of a circumferential free surface element, T_i is the average temperature of a free surface element, and T_∞ is the temperature of the back-radiation. In order to speed-up the calculation, the heat loss *via* radiation is simplified as a volumetric heat sink ξ_i in the free surface adjacent volume mesh elements

$$\xi_i = q_i / V_i \; [\mathrm{W/m^3}], \quad [2]$$

where V_i is the mesh element volume.

In the next step using ANSYS APDL, the distribution of harmonic EM field is calculated for the instant free surface shape of the electrode. Joule heat $Q(r_i, z_i)$ and Lorentz force density $f_r(r_i, z_i)$, $f_z(r_i, z_i)$ distributions, as well as previously calculated radiation heat sink $\xi(r_i, z_i)$ in the free surface adjacent mesh elements, are retrieved and interpolated back on the structured quadrilateral FLUENT mesh using User Defined Functions (UDFs).

The calculation of unsteady flow and heat transfer is performed further for sufficiently small time interval $\Delta t = 1$ ms using the updated heat and momentum sources. Time interval between recalculation of EM problem is considered sufficiently small if further decrease of Δt has no impact on the droplet detachment. Size of the FLUENT mesh element $\Delta x = 0.15$ mm has been chosen based on results of mesh studies aimed on finding a compromise between the higher computational accuracy and affordable computation time.

On the completion of flow calculation in FLUENT, a new instant free surface state is obtained and written into a file and the repeat of whole calculation loop ensures computation of free surface dynamics.

Details on the model implementation and extensive validation have been already presented previously.[12,13]

It was found that only the Scale Adaptive Simulation (SAS)[14] for the turbulence description combined with the VOF in 2D simplification was able to provide experimentally valid results without generation of overrated turbulent viscosity that distorts the modeling of droplet detachment.

The developed free surface simulation tool was upgraded by the heat transfer model including enthalpy-porosity technique for capturing the solidification/melting process and continuous casting approach for modeling downward movement of the electrode.[14]

The solid electrode enters the calculation domain with a constant velocity v_z at a temperature $T = 20$ °C through the boundary AB (Figure 3(b)). The electrode side surface BC moves downwards at the same velocity v_z and incorporates no-slip and radiation boundary conditions. Domain boundary CDEFG is a no-slip adiabatic wall and GH is an outlet. The convective cooling of the electrode by the gas flow is neglected since heat radiation is the dominant cooling mechanism.

III. RESULTS AND DISCUSSION

Induction melting calculations have been done using thermophysical properties summarized in Table I.

A. *Ø40 mm Tantalum Electrode*

Simulation results are presented for the fully developed melting of a Tantalum electrode (Ø40 mm, 30 cm length) using optimized induction coil, effective current $I_{\mathrm{ef}} = 2.6$ kA, and frequency $f = 186$ kHz. The new coil with less inductance ensured melting at $\times 1.5$ times higher frequency and achieved greater electrical efficiency since the total generator power was limited to only 50 kW.

Heat diffusion due to high thermal conductivity of Tantalum had to be compensated with a quite high-electrode feeding velocity (60 mm/min). This allowed for the lamellae-free melting of the electrode tip while the rest of the body remained cold, thus enhancing the melting stability and reducing the radiation loss from cylindrical side surface (Figure 4(a)).

The net Joule heat induced in the electrode is 24.3 kW. Radiation loss from cylindrical side surface is 1.2 kW (63 pct of power is emitted from 5 mm height cylindrical area right above the molten tip) and the power loss from the conical tip (droplets excluded) is 3.4 kW.

Induced eddy currents in the EM skin depth of the electrode are mainly azimuthal (j_φ) and closer to the axis their radius tends to zero. Therefore, Joule heat and Lorentz force have insignificant values in vicinity of the electrode axis (Figure 4(b)). This has to be considered to avoid re-solidification of the dripping melt especially at the start of the melting.

Surface tension-driven Rayleigh instability is the main physical mechanism responsible for the break-up of thin liquid metal stream at the tip of the electrode into smaller droplets. Generally, non-viscous infinite liquid column becomes unstable for perturbations with wavelengths λ greater than circumference $\pi \cdot D$, where D is the diameter of liquid cylinder. For viscous fluids, the fastest growing instability has the wave length[24]

$$\lambda = \sqrt{2}\pi D \sqrt{1 + 3\mu(\rho\gamma D)^{-1/2}} \quad [3]$$

and explains why thinner liquid streams break-up into smaller droplets.

Table I. Physical Properties of Pure Tantalum and Ti-6Al-4V Alloy

Property	Tantalum	Ti-6Al-4V
Density ρ, kg/m^3	16,650	4500
Solidus Temperature T_{sol}, °C	3016	1604
Liquidus Temperature T_{liq}, °C	3017	1650[20]
Latent Heat L, J/kg	202,100[15]	286,000[20]
Emissivity ε (Unoxidized, Near T_{liq})	0.25	0.3
Thermal Conductivity λ_{T}, W/(m K)	temp. dependent[16]	temp. dependent[21]
Specific Heat C_{p}, J/(kg K)	temp. dependent[17]	temp. dependent[22]
Viscosity μ, mPa s	8.0[18]	4.8[20]
Surface Tension γ, N/m	2.14[18]	1.52[23]
Electrical Resistivity ρ_{el}, 10^{-8} Ω m	130[19]	170[22]

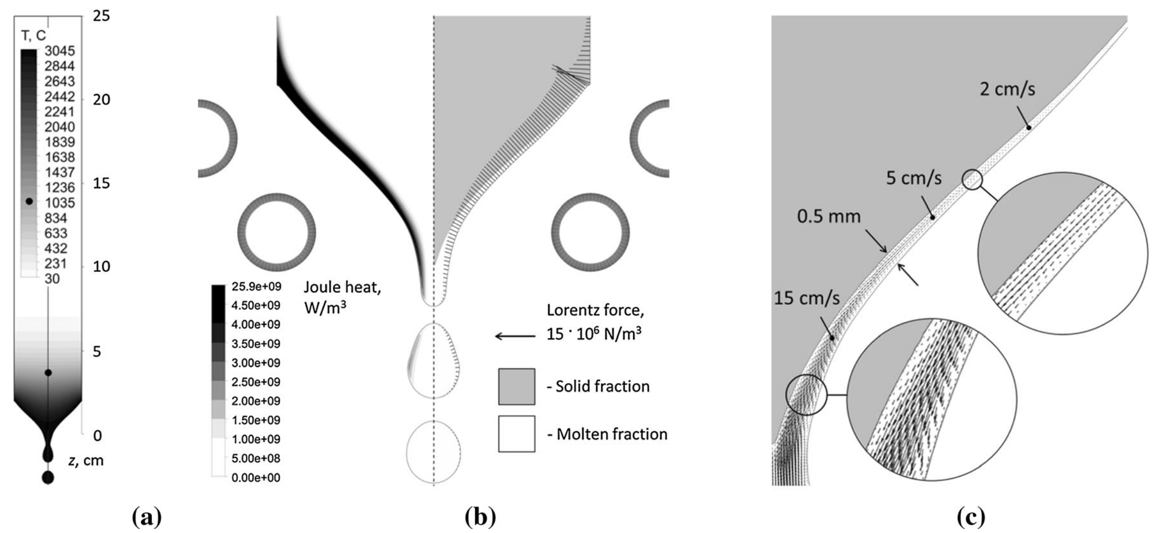


Fig. 4—(*a*) Temperature distribution in the electrode; (*b*) Joule heat (on the left), melting front, and Lorentz force vectors plotted at the free surface (on the right); (*c*) flow in the thin melt layer at the tip of the electrode.

The steady part of the Lorentz force (oscillating part is neglected here but can be implemented as well[25]) pinches the molten tip of the electrode, thus the coil design and melting parameters influence the diameter of the stream and therefore size of detached droplets. Moreover, maintaining the electrode melting rate and squeezing the stream will generate larger flow velocity and shift the break-up location closer to the nozzle. These factors play a crucial role in the final size of the atomized powder.

Apart from that, a method of droplet generation based on applying a modulated AC high-frequency magnetic field in the localized region of capillary break-up has been studied by other authors using numerical modeling and experiments.[26,27] It was found that the droplet size can be tailored by the frequency of the intermittent electromagnetic pinch force.

Under the gravity and Lorentz force pinching, a rapidly accelerating laminar flow is formed in the 0.5 mm thin melt film at the conical tip of the electrode (Figure 4(c)). Such small thickness of the molten layer does not allow to achieve high superheat, and the melt is being superheated only while it is accumulating below the tip of the electrode (Figure 5). Superheat of maximum 27 °C and velocity of 80 cm/s have been predicted by the model at the moment of droplet detachment. Because of the huge radiation loss, the falling droplet (diameter of 7.1 mm) loses almost all the superheat as it falls 3 cm below the lowest inductor winding. All experiences made with EIGA atomization processes have shown that the larger superheat of the atomized droplets favors formation of fine powders, therefore it is essential to mount the coil as close as possible to the atomization nozzle.

Maintaining the electrode feeding velocity v_z constant and increasing only the effective current of inductor I_{ef} will lead to the increase of induced power in the electrode. At this moment, the melting rate of the electrode will be faster than the electrode feeding rate. Because of that the distance between the conical electrode tip and the coil will start to increase until shortly the induced power will match again the feeding rate.

To illustrate the effect, the free surface and melting front shapes obtained for the fully developed melting of a Tantalum electrode at the reference and 1.23× increased inductor current are compared (Figure 6).

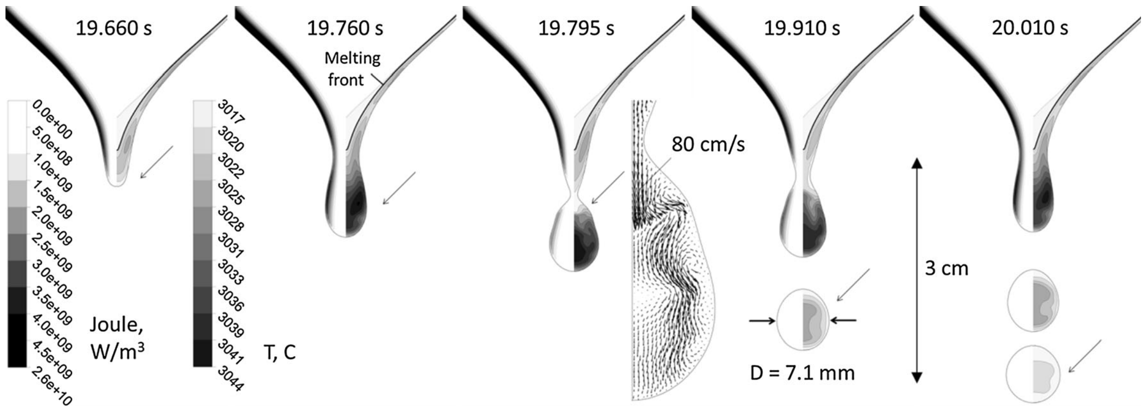


Fig. 5—Joule heat (on the left) and superheat above liquidus temperature T_{eiq} = 3017 °C (on the right) at different time moments of the simulated melt dripping from the tip of the Ø40 mm Tantalum electrode.

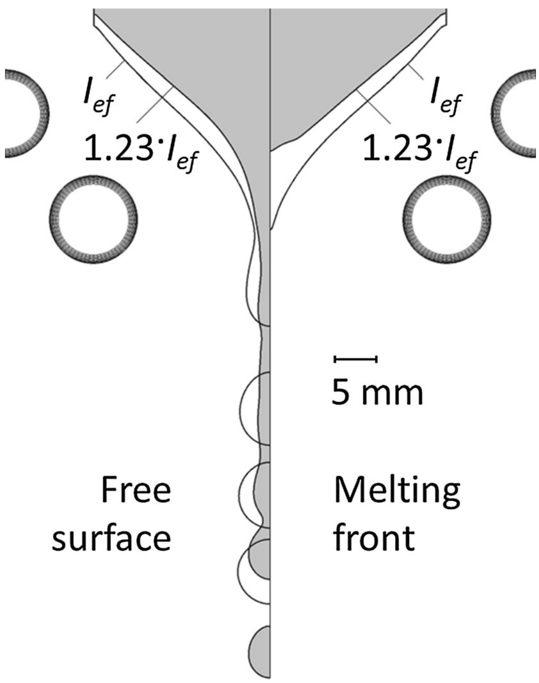


Fig. 6—Free surface (on the left) and melting front (on the right) shape comparison in case of different inductor current values I_{ef}.

At this new dynamic equilibrium with higher current, the magnetic field at the surface of the conical tip will be the same—increased induction will be compensated by increased distance to the coil. Meanwhile, the dripping melt has to pass through the center of the lowest winding where it will be inevitably exposed to a much stronger magnetic field due to higher current. This will lead to higher Lorentz force and Joule heat densities (Figure 7(a)) in this region (if EM skin-depth δ_{EM} is still smaller than the stream radius). The net Joule heat induced in the electrode using 1.23× higher current is 23.6 kW (only 3 pct smaller compared with the reference I_{ef} case).

Stronger EM pinching of the molten tip forces the melt to drain through a narrower cross-section while maintaining the same melting rate. Such EM funnel effect accelerates the melt stream that breaks up into droplets much lower if compared with less EM pinching. Therefore, note the different melting regimes: dripping of droplets in case of I_{ef} and continuous stream in case of $1.23 \cdot I_{\mathrm{ef}}$.

Larger Joule heat densities in the tip of the electrode lead to a prominent superheat with a maximum of 130 °C and move the melting front upward on the axis (Figure 7(b)). Further increase of inductor current and shift of the melting front lead to accumulation of molten metal below the electrode.

However, the bigger portion of molten metal below the electrode hinders process stability and is prone to fall as a large drop and clog the nozzle at the end of the melting when power is switched off. Therefore, it is recommended to finalize the melting by a smooth reduction of the power, allowing the melt to crystalize while still being confined by the Lorentz forces.

It has been observed that the low power and low AC frequency values lead to the growth of lamellae structures (Figure 2). Let us consider the horizontal (z = const) cross-section of the electrode tip (Figure 8). Because the real induction system is slightly non-axisymmetric (helicity of inductor, electrode misalignment, *etc.*) and the considered power is low, it is likely that instead of simultaneous melting of a circumferential ring at the tip of the electrode, the surface melting starts locally at $\varphi = \varphi_0$. The gravity and the Lorentz force push this small melt portion down and exposes the groove to EM field. Then two effects contribute to the deepening of the groove: eddy current density increasing at the bottom and decreasing at the edges of the groove and more intensive radiation cooling (bigger surface-to-volume ratio) at the edges of the groove. This also determines that the widening of the appeared groove is highly unlikely and the neighboring groove is more likely to appear at some distance d. Indeed, the distances between the grooves in tests appear to be comparable with EM skin-depth δ_{EM} that means that these lamellae structures become transparent in the EM

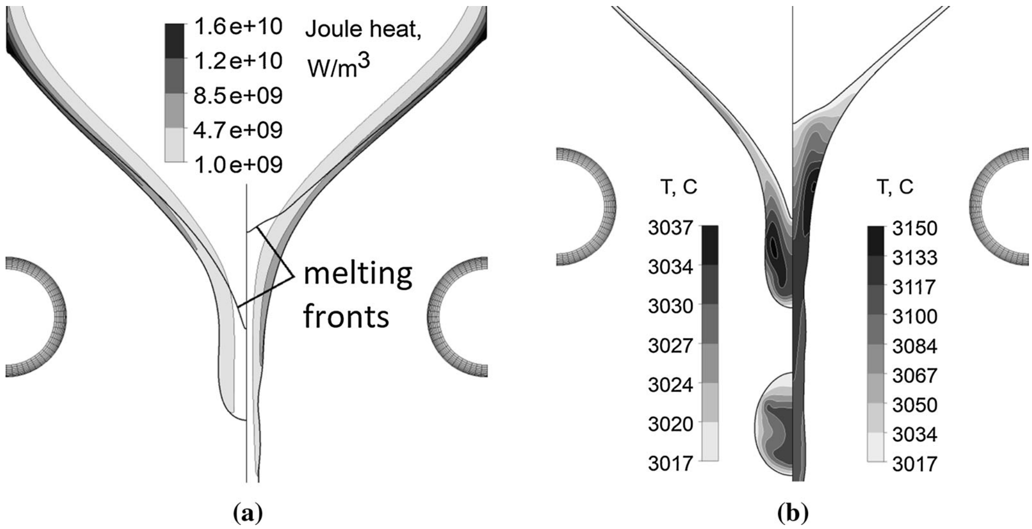


Fig. 7—Joule heat distribution (*a*) and molten metal superheat above liquidus temperature T_{liq} = 3017 °C (*b*) in the case of different inductor current values: I_{ef} (on the left) and $1.23{\cdot}I_{ef}$ (on the right).

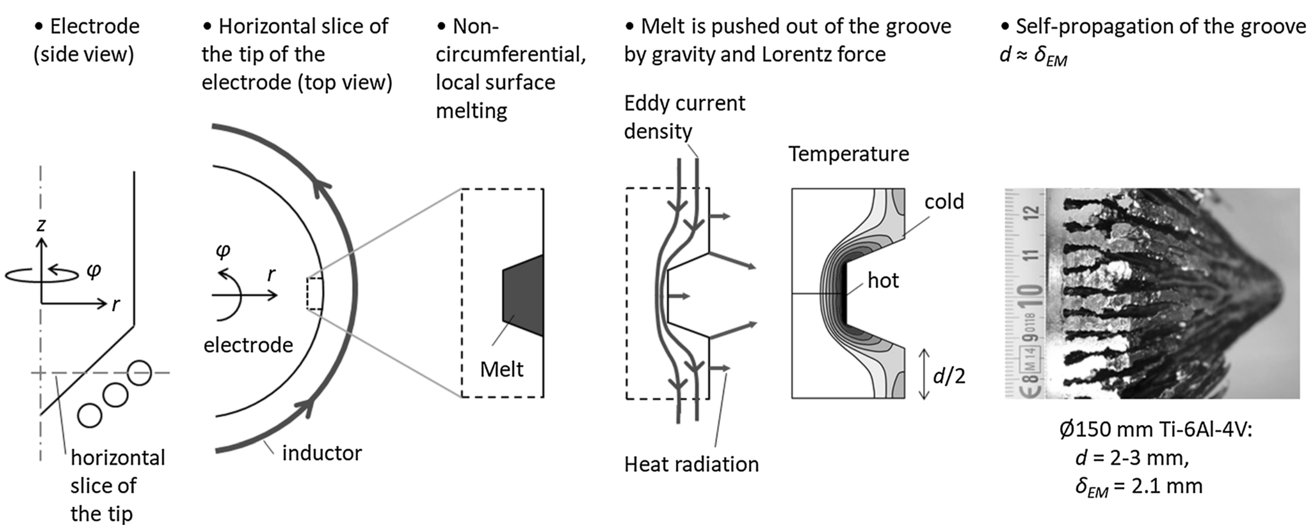


Fig. 8—Growth of lamellae structures at the tip of the electrode.

field and can no longer be melted inductively. In order to ensure coupling of these structures with EM field, one should prominently increase the AC frequency (*e.g.*, 10 times), however, the generators normally do not have such flexibility.

Using numerical modeling, the EIGA induction coil has been optimized and the process parameters have been tuned to meet the requirements for the stable, almost lamellae-free melting of the Ø40 mm Tantalum electrode at a melting rate of 1.25 kg/min. Note the fine agreement between the snapshot of the experiment (Figure 9(a)) and the simulation (Figure 9(b)).

Tantalum powder with mass-median-diameter d_{50} = 97 μm was obtained from Ø40 mm electrode during our first atomization tests. Successful Molybdenum (d_{50} = 84 μm) and Niobium (d_{50} = 65 μm) atomization trials with Ø50 mm electrodes have been done as well. Further optimization of melting conditions is being performed using the developed numerical model.

B. *Ø150 mm Ti-6Al-4V Electrode*

Optimization of the Ø150 mm Ti-6Al-4V electrode induction coil has been successful as well. Transient simulation using tuned process parameters, I_{ef} = 2.5 kA and f = 95 kHz predicted continuous stream steady-state melting regime. Insight in the melting process is illustrated by the Joule heat and Lorentz force distributions (Figure 10), as well as temperature and molten fraction contours (Figure 11).

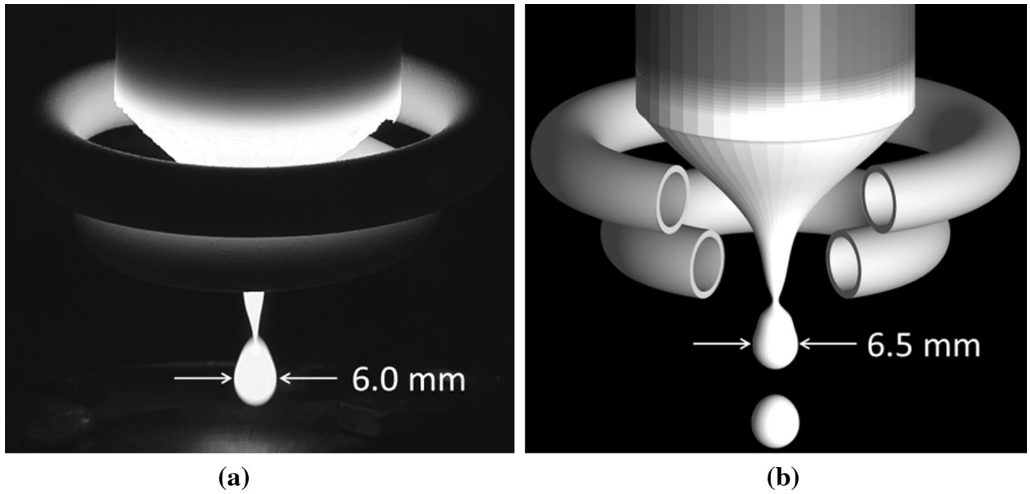


Fig. 9—Snapshot of Ø40 mm Tantalum electrode melting in (*a*) experiment and (*b*) simulation shortly before droplet detachment.

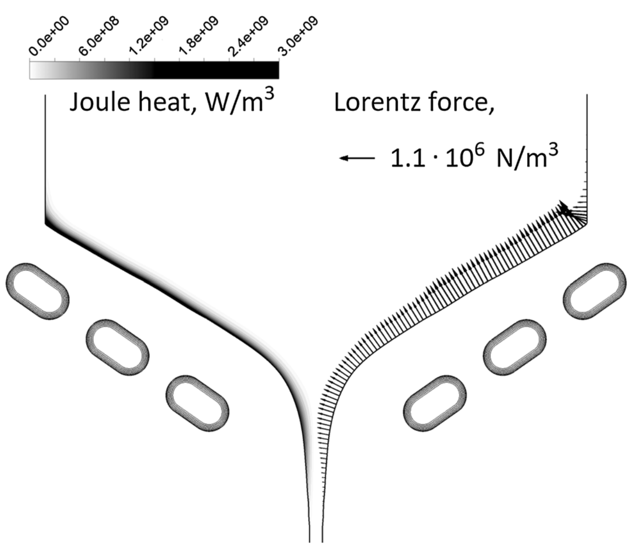


Fig. 10—Joule heat (on the left) and Lorentz force vectors plotted at the free surface of the electrode (on the right).

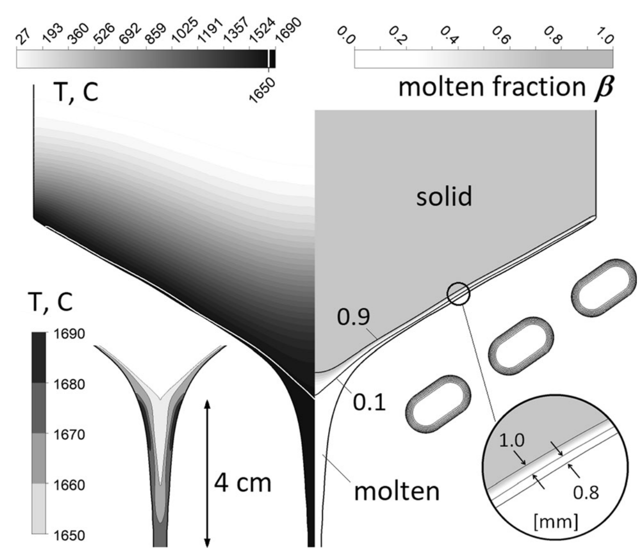


Fig. 11—Global temperature distribution and local superheat of the melt stream (on the left) and molten fraction (on the right).

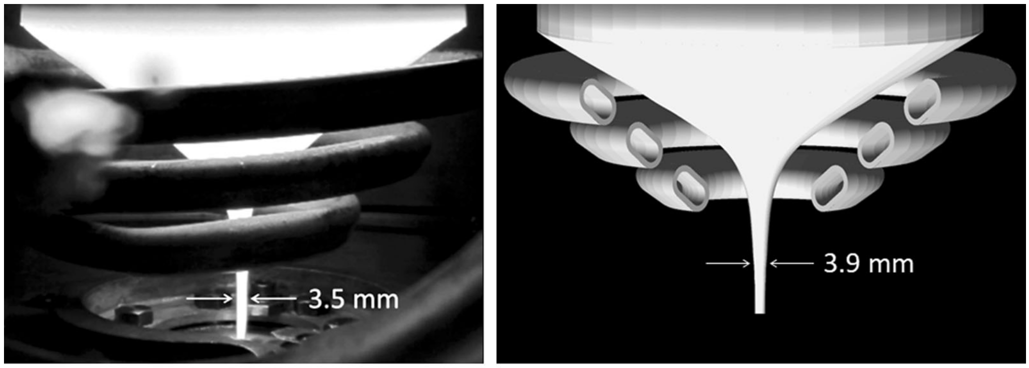
Low thermal conductivity of Ti-6Al-4V allows to maintain a steep temperature gradient along the axis of the electrode despite quite low electrode feeding velocity (30 mm/min). Radiation loss from the tip of the electrode is 5.2 kW while the power lost from cylindrical side surface is only 0.4 kW. The net Joule heat induced in the electrode is 60.9 kW.

A thin mushy layer (1.0 mm thickness) and fully liquid film (0.8 mm thickness) are formed at the conical solidus front of the electrode. Slight thickness of the liquid film does not allow for the high superheating and maximal temperature reaches only 5 to 10 K above liquidus temperature (T_{liq} = 1650 °C) at the free surface. The superheating takes place lower in the melt stream and the hotspot with maximum superheat of 40 K is located in the level of the lowest winding. Note the temperature inhomogeneity caused by the Joule heat distribution in the stream right below the tip of the solid electrode. According to simulation results, only 4 cm below the tip of the electrode the temperature is uniform and the melt stream has a superheat of 25 K.

Stable lamellae-free melting at a rate of 2 kg/min allowed for atomization of a thin continuous stream into a powder. Note the fine agreement between the snapshot of experiment (Figure 12(a)) and simulation (Figure 12(b)).

Ti-6Al-4V powder with mass-median-diameter d_{50} = 98 μm was obtained from Ø150 mm electrode during our first atomization trials. Further optimization of melting conditions is being performed using developed numerical model.

Further system scale-up is complicated by the coil's effective voltage limitation below 1000 V in order to avoid sparkling between uninsulated (ceramic-free) inductor windings. This constrains to design a coil with parallel windings that meanwhile delivers desired power distribution at the steady-state melting conical tip of the



(a) **(b)**

Fig. 12—Snapshot of Ø150 mm Ti-6Al-4V electrode melting using optimized induction coil and tuned process parameters in (*a*) experiment and (*b*) simulation.

electrode. In this case, the shape of the electrode tip has a tremendous impact on the current distribution in parallel windings that makes the problem quite complicated. Using the developed coupled simulation, we have designed a new induction coil that is capable of melting Ø250 mm Ti-6Al-4V electrode at a rate of 6 kg/min. Experimental validation of the new coil is the basis for further research.

IV. CONCLUSIONS

A 2D-axisymmetric model for the EIGA electrode induction melting has been developed. Fluid flow and heat transfer calculation in FLUENT are coupled to the Joule heat and Lorentz force recalculation upon the changing free surface shape in ANSYS APDL. The model is able to describe preheating, melting, formation of the melt stream at the tip of the electrode, droplet detachment, and steady-state dripping or streaming with detailed insight in the superheat distribution.

The comparison of our simulation results to experimental, semi-industrial observations revealed good agreement and validated the accuracy of developed approach.

Surface tension-driven Rayleigh instability is the main physical mechanism responsible for the break-up of thin liquid metal stream at the tip of the electrode into smaller droplets. Meanwhile, it can be influenced by inductor design and melting parameters.

Numerical modeling reveals that stronger EM pinching of the molten tip of the electrode (e.g., realized by increased inductor current) reduces diameter of the stream and therefore size of detached droplets. If the electrode feeding velocity is maintained, the squeezing of the stream will accelerate the flow and shift the break-up location closer to the nozzle.

In the steady-state condition, the electrode feeding rate matches the net induced power in the electrode. At this dynamic equilibrium with higher current, the magnetic field at the surface of the conical tip will be the same—increased induction will be compensated by increased distance between the electrode tip and the coil; however, the dripping melt has to pass through the center of the lowest winding where it will be inevitably exposed to a stronger EM field that will cause a larger superheat.

Thus, the coil design and melting parameters play a crucial role in the final size of the atomized powder.

By means of numerical modeling we were able to explain the problematic growth of lamellae structures at the tip of the electrode, tune ALD's EIGA systems for successful atomization of refractory materials with high melting temperatures like Tantalum (Ø40 mm), and ensure the further scale-up of already established processes (Ti-6Al-4V, Ø150 mm).

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